Lesson 1.1 **Duration: 40 minutes**

What is Networking?

Learning Objectives

- Define computer networking and its purpose
- Identify real-world examples of networks
- Explain how devices communicate through a network

Engage (5-7 minutes)

Imagine the bus route system in your town. Buses connect different areas—school, market, hospital—so people and messages can travel. A computer network works the same way: it connects computers so information can travel between them.

Explore (10-12 minutes)

Draw a simple diagram of your school computer lab. How many computers are there? Are they connected to each other or to a printer? Write down what you think happens when you print a document from one computer.

Explain (10-12 minutes)

A computer network is a group of two or more computers connected together to share data, files, and resources. Networks can be small, like two computers in a room, or huge, like the internet connecting the whole world. The main purpose of networking is communication and resource sharing.

Elaborate (8-10 minutes)

Think of your hostel or home. If two siblings want to share a photo, they might pass a phone around. Now imagine five computers connected—they can share the same printer, internet connection, and files instantly without passing anything around. Name three things a network could share in your classroom.

Evaluate (5-8 minutes)

Exit ticket: In one sentence, define what a computer network is. Then name one thing a network helps us share. If you finish early, draw a picture of two computers connected by a cable.

Lesson 1.2 **Duration: 40 minutes**

Types of Networks

Learning Objectives

- Differentiate between LAN, MAN, and WAN
- Give examples of each network type from daily life
- Understand the range and speed differences between network types

Engage (5-7 minutes)

Think about three things: your classroom LAN cable connecting all computers, the cable TV network covering your whole town, and the internet connecting India to the world. These are like local, city-wide, and worldwide roads for data—each covering a different distance.

Explore (10-12 minutes)

Match these network types to their coverage: LAN, MAN, WAN. Draw three circles representing small, medium, and large areas and label each with the correct network type. Discuss with a partner why your school likely uses a LAN.

Explain (10-12 minutes)

LAN (Local Area Network) connects computers in a small area like a school or office building. MAN (Metropolitan Area Network) covers a city or large campus, like a cable TV network. WAN (Wide Area Network) spans countries and continents—the internet is the largest WAN in the world.

Elaborate (8-10 minutes)

Consider a village panchayat setup. If five computers in the panchayat office share a printer, that is a LAN. If the panchayat office connects to the block office ten kilometers away, that is a MAN. If the block office connects to the state capital, that is a WAN. Which type would connect your school to the district education office?

Evaluate (5-8 minutes)

Exit ticket: Write the full form of LAN, MAN, and WAN. Then give one real-life example of each. If you finish early, explain why a WAN is slower than a LAN.

Lesson 1.3 **Duration: 40 minutes**

Network Devices

Learning Objectives

- Identify common network devices and their functions
- Explain the role of routers, switches, and modems
- Differentiate between wired and wireless devices

Engage (5-7 minutes)

Picture a railway junction. Trains arrive from different cities, and the station master directs each train to the correct platform. Network devices work the same way—they receive data and route it to the right destination. Today we meet the station masters of the computer world.

Explore (10-12 minutes)

Look at the networking corner of your computer lab. Can you find a modem, router, or switch? Draw each device you see and write one sentence describing what you think it does. Compare your drawings with a classmate.

Explain (10-12 minutes)

A modem converts data between your network and the internet. A router directs data packets to the correct devices on a network, like a postmaster sorting letters. A switch connects multiple computers within a LAN so they can communicate. A hub is an older device that sends data to all ports regardless of destination.

Elaborate (8-10 minutes)

Think of the bus stand in your town. The modem is like the bus that travels between towns carrying passengers. The router is the bus stand conductor who directs passengers to the right bus. The switch is like the seating arrangement inside the bus that lets passengers talk to their neighbors. Which device would you need to connect two computers in a small office?

Evaluate (5-8 minutes)

Exit ticket: Name three network devices and write one function for each. If you finish early, explain why a switch is better than a hub for a school network.

Lesson 1.4 **Duration: 40 minutes**

Internet Connection

Learning Objectives

- Identify different types of internet connections
- Compare broadband, mobile data, and Wi-Fi
- Understand the concept of bandwidth and speed

Engage (5-7 minutes)

Think about the water supply in your village. Some houses get water through pipes, others carry it in pots from a well, and some use tankers. Internet connections are similar—broadband is like piped water, mobile data is like carrying a pot, and Wi-Fi is like a community tap. Each has its own capacity and convenience.

Explore (10-12 minutes)

List all the ways you or your family access the internet—mobile phone, computer lab, home Wi-Fi. Write the name of the service provider for each. Discuss with a partner which connection feels fastest and why.

Explain (10-12 minutes)

Broadband provides a fast, constant internet connection through cables or DSL. Mobile data (3G, 4G, 5G) uses cellular towers to connect your phone anywhere. Wi-Fi is a wireless technology that connects devices to a router within a limited range. Bandwidth measures how much data can flow per second, like the width of a water pipe.

Elaborate (8-10 minutes)

In your village, different families use different water sources. Similarly, different people use different internet sources. A student watching online classes at home might use mobile data, while the school lab uses broadband. If the school broadband slows down, it is like a narrow pipe during peak hours. Name one advantage and one disadvantage of mobile data over broadband.

Evaluate (5-8 minutes)

Exit ticket: Name two types of internet connections and write one difference between them. If you finish early, explain what bandwidth means using a water pipe analogy.

Lesson 1.5 **Duration: 40 minutes**

Internet Protocols

Learning Objectives

- Define what an internet protocol is
- Explain the role of IP addresses and URLs
- Understand how HTTP and HTTPS work

Engage (5-7 minutes)

When you send a letter, it needs an address, a pin code, and a postal system to deliver it. The internet works the same way—every device has an address (IP), every website has a name (URL), and protocols like HTTP are the postal rules that deliver web pages to your screen.

Explore (10-12 minutes)

Type a website address like www.google.com in a browser. Look at the address bar—what do you see? Write down the full URL. Now ask your teacher to show you an IP address using a command. Compare the URL and IP address.

Explain (10-12 minutes)

An IP address is a unique number assigned to every device on a network, like a house number on a street. A URL is the human-readable name of a website, like www.example.com. HTTP (HyperText Transfer Protocol) sends web pages without encryption, while HTTPS adds security by encrypting data—like sending a letter in a sealed envelope versus a postcard.

Elaborate (8-10 minutes)

Think of the panchayat office. Each office has a unique address for mail delivery. Similarly, each computer on the internet has an IP address. The URL is like the office name—"Gram Panchayat Office"—while the IP is like its GPS coordinates. Why do you think HTTPS is important when banking online?

Evaluate (5-8 minutes)

Exit ticket: Write the difference between an IP address and a URL. Then explain why HTTPS is safer than HTTP. If you finish early, look up the IP address of any website using an online tool.

Lesson 1.6	Duration: 40 minutes

Email Services

Learning Objectives

- Understand what email is and how it works
- Identify the parts of an email address
- Compose a basic email with subject, body, and recipient

Engage (5-7 minutes)

Imagine you want to send a message to your friend in another village. You could walk there, send a letter by post, or use a phone. Email is like sending an instant letter through the internet—it reaches your friend in seconds, no matter how far away they are. Today we learn how to send these electronic letters.

Explore (10-12 minutes)

Open an email application or webmail. Look at your own email address—what parts do you see before and after the @ symbol? Write down the structure: username @ domain. Then open a received email and identify the sender, subject, and body.

Explain (10-12 minutes)

Email stands for electronic mail. An email address has two parts: the username (your name or ID) and the domain (the service provider like gmail.com). Every email needs a recipient address, a subject line describing the content, and the body containing the message. Emails can also include attachments like documents or photos.

Elaborate (8-10 minutes)

Think of sending a letter through India Post. You write the address (recipient), a subject on the envelope (like “Urgent—Examination Schedule”), and the letter inside (body). Email works identically but electronically. Your teacher might email your parents—what subject line would be most helpful for them to read first?

Evaluate (5-8 minutes)

Exit ticket: Write the three main parts of an email. Then draft a one-sentence email to your teacher asking for homework help. If you finish early, explain the difference between CC and BCC.

Lesson 1.7 **Duration: 40 minutes**

Attachments

Learning Objectives

- Attach files to an email correctly
- Understand file size limits and common formats
- Open and save email attachments safely

Engage (5-7 minutes)

When you send a letter through the post, you can include a photograph or a document inside the envelope. An email attachment works the same way—it lets you send files like images, documents, or presentations along with your message. Today we learn how to pack these digital parcels.

Explore (10-12 minutes)

Open your email and start composing a new message. Find the attachment button (usually a paperclip icon). Try attaching a small image or text file from your computer. Note the file size—does it show a limit? Discuss what happens if the file is too large.

Explain (10-12 minutes)

An attachment is a file added to an email. Common formats include documents (.doc, .pdf), images (.jpg, .png), and presentations (.ppt). Most email services limit attachment size to 25 MB. Always scan attachments before opening them to avoid viruses. You can download attachments by clicking the download button.

Elaborate (8-10 minutes)

Think of sending a parcel through a courier. There is a weight limit, you must pack it properly, and the receiver should check the parcel before opening it. Similarly, email attachments have size limits, should be in common formats, and recipients should verify the sender before downloading. Why might a school assignment submission require a PDF instead of a Word file?

Evaluate (5-8 minutes)

Exit ticket: Name two common attachment formats and explain one safety rule for handling attachments. If you finish early, practice attaching and sending a file to a classmate's email.

Lesson 1.8 **Duration: 40 minutes**

Internet Safety

Learning Objectives

- Identify common online threats and scams
- Practice safe browsing habits
- Understand the importance of strong passwords

Engage (5-7 minutes)

When you walk through a busy market, you watch for pickpockets and strangers offering too-good deals. The internet is like that crowded market—full of useful things but also scams, fake links, and strangers who might not be who they say they are. Today we learn how to stay safe in this digital bazaar.

Explore (10-12 minutes)

Look at these scenarios: a pop-up saying “You won a prize! Click here,” an email asking for your bank password, and a friend request from someone you do not know. Which ones are suspicious? Discuss with a partner why each might be dangerous and what you should do.

Explain (10-12 minutes)

Internet safety means protecting yourself from online threats. Never share personal information like passwords, address, or phone number with strangers. Use strong passwords with letters, numbers, and symbols. Avoid clicking on unknown links or downloading files from untrusted sources. Always verify suspicious messages before responding.

Elaborate (8-10 minutes)

In your village, you would not give your house key to a stranger. Similarly, you should never share your passwords online. A strong password is like a sturdy lock on your door—it keeps intruders out. Think of three things you should never share online and explain why each is dangerous.

Evaluate (5-8 minutes)

Exit ticket: Write three rules for staying safe on the internet. Then create a strong password using letters, numbers, and symbols. If you finish early, list two signs that an email might be a phishing scam.

Lesson 1.9 **Duration: 40 minutes**

Social Media

Learning Objectives

- Identify popular social media platforms
- Understand the benefits and risks of social media
- Practice responsible social media behavior

Engage (5-7 minutes)

Social media is like the village square where everyone gathers to chat, share news, and celebrate festivals. Platforms like WhatsApp, YouTube, and Instagram are digital village squares—but unlike the real one, anyone in the world can join. Today we learn how to use these spaces wisely.

Explore (10-12 minutes)

List all social media apps or websites you or your family use. Write one positive thing and one negative thing about each. For example: WhatsApp helps stay connected but can spread fake news. Share your list with the class.

Explain (10-12 minutes)

Social media platforms like WhatsApp, Facebook, Instagram, and YouTube let people share photos, videos, and messages. Benefits include staying connected with family, learning new things, and expressing creativity. Risks include cyberbullying, privacy invasion, and addiction. Always think before you post—once something is online, it is very hard to remove.

Elaborate (8-10 minutes)

Think of the village notice board. Anyone can post a message, but false information causes problems. Social media is the same—a wrong post can spread to thousands in minutes. Before posting, ask yourself: Is it true? Is it kind? Is it necessary? Name one situation where social media helped your community.

Evaluate (5-8 minutes)

Exit ticket: Write two benefits and two risks of social media. Then list three questions you should ask yourself before posting online. If you finish early, write a short paragraph about responsible social media use.

Lesson 1.10 **Duration: 40 minutes**

Download Management

Learning Objectives

- Download files safely from the internet
- Understand file types and where downloads are saved
- Organize and manage downloaded files

Engage (5-7 minutes)

When you buy supplies from the market, you bring them home and store them in the right place—grains in the kitchen, books on the shelf. Downloading files works the same way. You bring digital files from the internet to your computer and need to store them properly so you can find them later.

Explore (10-12 minutes)

Open a browser and search for a free image or document. Click the download button and notice where the file saves by default. Check your Downloads folder—how many files are there? Try renaming and moving one file to a different folder.

Explain (10-12 minutes)

Downloading means copying a file from the internet to your computer. Files save in the Downloads folder by default. You can change the save location before downloading. Always check file types (.pdf, .jpg, .exe) before opening—executable files (.exe) can contain viruses. Keep your downloads folder organized by creating subfolders for different file types.

Elaborate (8-10 minutes)

Think of your school bag. If you throw everything inside—books, lunch box, pencil case—finding anything becomes hard. The Downloads folder is like an unorganized bag. Create folders for "Images," "Documents," and "Presentations" to keep things tidy. Why is it important to know the file type before opening a download?

Evaluate (5-8 minutes)

Exit ticket: Name two safety tips for downloading files. Then describe how you would organize your Downloads folder. If you finish early, practice downloading a file and moving it to a new folder.

Lesson 1.11 **Duration: 40 minutes**

Network Troubleshooting

Learning Objectives

- Identify common network problems
- Use basic troubleshooting steps
- Understand when to seek technical help

Engage (5-7 minutes)

When the water supply stops in your house, you check the tap, then the pipe, then call the plumber. Network problems are similar—first check the simple things like cables and Wi-Fi, then move to bigger issues. Today we learn how to become our own network plumbers.

Explore (10-12 minutes)

Your teacher will simulate a network problem—disconnect a cable or turn off Wi-Fi. Try to find the problem using these steps: Is the cable plugged in? Is Wi-Fi turned on? Can you ping another computer? Write down what you find at each step.

Explain (10-12 minutes)

Common network problems include loose cables, disabled Wi-Fi, incorrect passwords, and router issues. Basic troubleshooting steps are: check physical connections, restart your computer and router, verify network settings, and test with another device. If the problem persists, contact your network administrator or ISP.

Elaborate (8-10 minutes)

Think of fixing a bicycle. First check if the tire is flat, then the chain, then the brakes—start simple before assuming the worst. Network troubleshooting follows the same logic: check the obvious first. If your computer cannot connect to Wi-Fi, what are the first three things you should check?

Evaluate (5-8 minutes)

Exit ticket: List four steps to troubleshoot a network connection problem. Then write one situation where you would need to call for technical help. If you finish early, practice the ping command to test a connection.

Lesson 1.12 **Duration: 40 minutes**

Chapter 1 Review

Learning Objectives

- Review all Chapter 1 networking concepts
- Connect networking topics to real-world applications
- Identify areas needing further practice

Engage (5-7 minutes)

Think of a village fair where you learned about different stalls—food, games, education. Today we revisit all the networking stalls we explored: what networks are, their types, devices, internet connections, protocols, email, safety, and troubleshooting. Let us see how much you remember.

Explore (10-12 minutes)

Complete this networking crossword: the answers include LAN, WAN, router, modem, IP address, HTTPS, email, attachment, password, and bandwidth. Work in pairs to fill it in as fast as possible. Check your answers against the word bank provided by your teacher.

Explain (10-12 minutes)

Chapter 1 covered networking fundamentals: what networks are, types like LAN/MAN/WAN, devices like routers and modems, internet connections, protocols, email services, attachments, internet safety, social media, downloads, and troubleshooting. Each concept builds on the previous one to form a complete picture of how computers communicate.

Elaborate (8-10 minutes)

Now that you know about networking, think about your village. If the panchayat wanted to set up a network connecting the school, hospital, and office, what would they need? List the devices, connection type, and safety measures. This project thinking helps you apply everything you learned.

Evaluate (5-8 minutes)

Chapter quiz: Answer ten questions covering all topics from Lesson 1.1 to 1.11. Include definitions, examples, and short-answer questions. If you finish early, help a classmate with any questions they find difficult.

Lesson 2.1 **Duration: 40 minutes**

Introduction to GIMP

Learning Objectives

- Open and navigate the GIMP interface
- Identify key toolbox elements and menus
- Create and save a new image file

Engage (5-7 minutes)

Think of a village artist who paints banners for festivals. They use brushes, colors, a palette, and a canvas. GIMP is like a digital artist's studio—it has brushes, colors, layers, and a canvas on your computer screen. Today we enter this powerful free painting studio.

Explore (10-12 minutes)

Open GIMP on your computer. Explore the interface: the toolbox on the left, the canvas in the center, and the layers panel on the right. Click on different tools and see what happens. Try creating a new file and setting the canvas size to 800 by 600 pixels.

Explain (10-12 minutes)

GIMP (GNU Image Manipulation Program) is a free, open-source image editor similar to Photoshop. The toolbox contains selection tools, paint brushes, text tools, and more. The canvas is your workspace where you create and edit images. The layers panel helps you organize different parts of your image. Always save your work in GIMP's native format (.xcf) to preserve layers.

Elaborate (8-10 minutes)

Imagine setting up a rangoli design. You start with a base outline, add colors layer by layer, and adjust details. GIMP works the same way—you build your image using different tools and layers. What would you create if you had a digital canvas: a poster for your school, a greeting card, or a photo edit?

Evaluate (5-8 minutes)

Exit ticket: Name three parts of the GIMP interface and their functions. Then save a blank image as "practice.xcf" in your folder. If you finish early, try using the pencil tool to draw a simple shape.

Lesson 2.2 **Duration: 40 minutes**

Selection Tools

Learning Objectives

- Use rectangle, ellipse, and free select tools
- Understand when to use each selection type
- Practice selecting and modifying portions of an image

Engage (5-7 minutes)

When a tailor cuts cloth, they first draw the outline of the piece they need. Selection tools in GIMP are like that tailor's chalk—they let you mark exactly which part of the image you want to change, move, or delete. Today we learn to outline the parts we want to work with.

Explore (10-12 minutes)

Open a sample image in GIMP. Try the Rectangle Select tool—drag a box around part of the image. Then try the Ellipse Select tool for a circular selection. Finally, try the Free Select (lasso) tool to draw a custom shape. Copy and paste each selection to see the result.

Explain (10-12 minutes)

Selection tools define which area of an image you want to edit. Rectangle Select creates square or rectangular selections. Ellipse Select creates circular or oval selections. Free Select (Lasso) lets you draw any shape by hand. After selecting, you can cut, copy, fill, or apply effects only to the selected area.

Elaborate (8-10 minutes)

Think of drawing a boundary around a field for planting crops. The rectangle is like a paddy field, the ellipse is like a circular garden bed, and the free select is like an irregular plot along a riverbank. Each shape suits different needs. Which selection tool would you use to isolate a person's face from a group photo?

Evaluate (5-8 minutes)

Exit ticket: Name three selection tools and give one use case for each. Then select a portion of an image and change its color. If you finish early, practice using the feather option to soften selection edges.

Lesson 2.3 **Duration: 40 minutes**

Layers Concept

Learning Objectives

- Understand what layers are and why they matter
- Create, rename, and reorder layers
- Use layer visibility and opacity controls

Engage (5-7 minutes)

Imagine making a sandwich. Bread is the bottom layer, then vegetables, then cheese, then another bread on top. Each layer is separate but together they make the final sandwich. In GIMP, layers work exactly like that—each element sits on its own transparent sheet, stacked to create the final image.

Explore (10-12 minutes)

Open GIMP and create a new image. Add a new layer using the Layers panel. Draw something on the background layer, then switch to the new layer and draw something different. Toggle the eye icon to show and hide each layer. Adjust the opacity slider and observe the effect.

Explain (10-12 minutes)

Layers are transparent sheets stacked on top of each other. Each layer holds separate parts of an image, so you can edit one without affecting others. The background layer is the bottom sheet. You can rename layers for organization, change their order, adjust opacity (transparency), and toggle visibility. Layers are stored from bottom to top in the Layers panel.

Elaborate (8-10 minutes)

Think of making a puppet show. The background is a painted village scene, characters are cutouts on sticks moved in front, and effects like rain are 透明 overlays. Each is a separate layer. If you want to change the background without touching the characters, you simply edit the background layer. Why might an artist prefer many small layers over one flat image?

Evaluate (5-8 minutes)

Exit ticket: Explain what a layer is using the sandwich analogy. Then create three layers, name them, and change their order. If you finish early, practice adjusting layer opacity to create a see-through effect.

Lesson 2.4 **Duration: 40 minutes**

Layer Effects

Learning Objectives

- Apply drop shadows and glows to layers
- Use layer masks for non-destructive editing
- Combine multiple layer effects for creative results

Engage (5-7 minutes)

When the sun shines behind a tree, the tree casts a shadow on the ground. Layer effects in GIMP add similar shadows, glows, and depth to your images. Today we learn how to make our digital artwork look three-dimensional and professional.

Explore (10-12 minutes)

Open an image with text or shapes on a separate layer. Right-click the layer and explore "Layer Effects" or "Filters > Light and Shadow." Apply a drop shadow to your text. Adjust the shadow distance, blur, and color. Try adding an outer glow effect to a shape.

Explain (10-12 minutes)

Layer effects add visual depth and style to layers. Drop Shadow creates a shadow behind a layer, making it appear lifted. Outer Glow adds a colored aura around an object. Bevel and Emboss give a 3D raised appearance. Layer masks hide parts of a layer without permanently erasing them—paint black on the mask to hide, white to reveal.

Elaborate (8-10 minutes)

Think of decorating a school notice board. You pin papers (layers) and add borders, shadows, and highlights to make them stand out. Drop shadow is like the shadow a pinned paper casts on the board. Layer masks are like using removable tape—you can hide or reveal parts without cutting the paper. What effect would you add to a Diwali greeting card?

Evaluate (5-8 minutes)

Exit ticket: Name two layer effects and describe what each does. Then apply a drop shadow to text in GIMP. If you finish early, experiment with layer masks to blend two images together.

Lesson 2.5 **Duration: 40 minutes**

Drawing Tools

Learning Objectives

- Use pencil, paintbrush, and airbrush tools
- Adjust brush size, shape, and opacity
- Create freehand drawings and simple illustrations

Engage (5-7 minutes)

A village artist uses different brushes—a thin brush for fine details, a wide brush for filling backgrounds, and a spray can for soft shading. GIMP's drawing tools work the same way: pencil for sharp lines, paintbrush for smooth strokes, and airbrush for gentle gradients. Today we become digital artists.

Explore (10-12 minutes)

Select the Pencil tool and draw a line. Then switch to the Paintbrush and draw the same line—notice the difference in edge softness. Try the Airbrush tool and adjust the rate slider. Change brush sizes using the bracket keys [and]. Draw a simple house with trees using all three tools.

Explain (10-12 minutes)

The Pencil tool draws hard-edged lines with no anti-aliasing, good for pixel art. The Paintbrush draws smooth, anti-aliased strokes that blend naturally. The Airbrush applies color gradually like a real spray, building up intensity over time. All tools allow adjusting size, opacity, and flow to control how much color is applied.

Elaborate (8-10 minutes)

Think of rangoli making. The pencil is like using a single grain of rice for fine outlines. The paintbrush is like using colored powder for filling shapes. The airbrush is like gently sprinkling powder for a soft, blended effect. Each tool serves a different purpose in creating the full picture. Which tool would you use to draw the outline of a festival banner?

Evaluate (5-8 minutes)

Exit ticket: Describe one difference between the pencil and paintbrush tools. Then draw a simple scene using at least two different brushes. If you finish early, experiment with brush dynamics and opacity settings.

Lesson 2.6 **Duration: 40 minutes**

Color Adjustments

Learning Objectives

- Adjust brightness, contrast, and color balance
- Use hue-saturation to modify colors
- Correct common photo problems like underexposure

Engage (5-7 minutes)

Have you seen a photo taken at sunset where everything looks orange? Or a dark photo where you cannot see faces clearly? Color adjustments fix these problems—like adjusting the brightness of a bulb or the tint of a window. Today we learn to make every photo look its best.

Explore (10-12 minutes)

Open a dark or blurry photo in GIMP. Go to Colors > Brightness-Contrast and increase brightness. Then try Colors > Hue-Saturation to make colors more vibrant. Finally, try Colors > Color Balance to warm up or cool down the image. Save before and after versions to compare.

Explain (10-12 minutes)

Brightness controls how light or dark an image is. Contrast adjusts the difference between light and dark areas. Hue-Saturation changes the color intensity and shade. Color Balance shifts the mix of red, green, and blue tones. These adjustments help fix photos that are too dark, too bright, or have unwanted color casts.

Elaborate (8-10 minutes)

Think of adjusting the lights in your house. Too bright and you cannot see the screen; too dark and you cannot read. Photo adjustment is like finding that perfect lighting. If a photo from a school function came out too yellow due to indoor lights, which adjustment would fix it?

Evaluate (5-8 minutes)

Exit ticket: Name two color adjustment tools and describe what each does. Then adjust a photo's brightness and contrast. If you finish early, try correcting a color cast using Color Balance.

Lesson 2.7 **Duration: 40 minutes**

Filters Introduction

Learning Objectives

- Understand what filters do to images
- Apply blur, sharpen, and noise filters
- Use the preview feature before applying filters

Engage (5-7 minutes)

When you look through a frosted glass window, the world outside looks blurry. Filters in GIMP are like different types of glasses—they transform how an image looks. Some make it sharper, some softer, some add artistic effects. Today we try on different digital glasses.

Explore (10-12 minutes)

Open an image in GIMP. Go to Filters > Blur and try Gaussian Blur—adjust the blur radius. Then try Filters > Enhance > Sharpen. Compare before and after using the preview checkbox. Try adding noise with Filters > Noise and observe the effect.

Explain (10-12 minutes)

Filters apply mathematical transformations to image pixels. Blur filters soften images by averaging nearby pixels. Sharpen filters increase contrast between adjacent pixels to make details stand out. Noise filters add random grain for texture or artistic effect. Always use the preview option before applying a filter to see the result, and use Undo (Ctrl+Z) if you do not like it.

Elaborate (8-10 minutes)

Think of looking through different types of glass—a clear window (no filter), frosted glass (blur), magnifying glass (sharpen), and textured glass (noise). Each changes how you see the world. Filters do the same to images. Why would you use blur on a background but sharpen on a portrait?

Evaluate (5-8 minutes)

Exit ticket: Name three types of filters and describe the effect of each. Then apply a blur filter to an image and undo it. If you finish early, try the Pixelize filter for a mosaic effect.

Lesson 2.8 **Duration: 40 minutes**

Artistic Filters

Learning Objectives

- Apply oil painting and watercolor effects
- Use the cartoon filter for stylized images
- Combine multiple artistic filters creatively

Engage (5-7 minutes)

Your art teacher uses different painting styles—watercolor for soft washes, oil paint for rich textures, and pencil sketches for outlines. GIMP has artistic filters that mimic these same styles. Today we transform photographs into works of art using digital painting techniques.

Explore (10-12 minutes)

Open a photograph in GIMP. Go to Filters > Artistic and try Oilify. Adjust the mask size slider and preview the effect. Then try the Waterpixels filter for a watercolor look. Try the Cartoony filter for a comic-book style. Save each version with a different filename.

Explain (10-12 minutes)

Artistic filters transform photos to look like traditional art. Oilify simulates oil painting by blending colors in brushstroke patterns. Waterpixels creates a watercolor effect with soft edges. The Cartoon filter adds bold outlines and flat colors like a comic strip. These filters work best on high-resolution images and can be combined for unique effects.

Elaborate (8-10 minutes)

Think of the Warli paintings on village walls. Each style has its own character—oil painting looks rich, watercolor looks delicate, and cartoon looks fun. Artistic filters let you recreate these styles digitally. If you wanted to turn a photo of your school into a Warli-style painting, which approach would you take?

Evaluate (5-8 minutes)

Exit ticket: Name two artistic filters and describe the effect each produces. Then apply an oil painting filter to a photo. If you finish early, combine blur and artistic filters for a unique effect.

Lesson 2.9 **Duration: 40 minutes**

Text and Typography

Learning Objectives

- Add and format text on images
- Change font, size, color, and alignment
- Apply text effects like shadows and outlines

Engage (5-7 minutes)

Think of the colorful posters outside your school during annual day. Each poster has big titles, small subtitles, and different fonts that catch your eye. In GIMP, the Text tool lets you add these same eye-catching words to any image. Today we become poster designers.

Explore (10-12 minutes)

Select the Text tool (A icon) and click on your image. Type your school name. Highlight the text and explore the options: change font to something decorative, increase size to 72, change color to red, and try center alignment. Add a second text block with different formatting.

Explain (10-12 minutes)

The Text tool adds editable text layers to images. You can change font family, size, color, alignment (left, center, right), and spacing. Text in GIMP remains editable until you rasterize it. For effects, right-click the text layer and add a drop shadow or outline using Layer Effects. Use contrasting colors for readability.

Elaborate (8-10 minutes)

Think of writing on a blackboard. You choose chalk color, write big for titles and small for details, and center important text. The Text tool works similarly. A school poster needs a large title, medium subtitle, and small details. Practice creating a poster with three text levels. What colors would you choose for a science fair poster?

Evaluate (5-8 minutes)

Exit ticket: List three text formatting options available in GIMP. Then create a title with a shadow effect. If you finish early, design a simple event poster with multiple text styles.

Lesson 2.10 **Duration: 40 minutes**

Image Composition

Learning Objectives

- Combine multiple images into one composition
- Use cropping and resizing for proper layout
- Apply the rule of thirds for balanced design

Engage (5-7 minutes)

When you create a collage for a school project, you cut pictures from magazines, arrange them on a chart paper, and add borders and labels. Image composition in GIMP is the same process but digital—you combine photos, adjust sizes, and arrange elements for a professional result.

Explore (10-12 minutes)

Open two different images in GIMP. Select part of one image, copy it, and paste it as a new layer onto the second image. Use the Scale tool to resize the pasted image. Then try the Crop tool to trim unwanted areas. Arrange the elements so the composition looks balanced.

Explain (10-12 minutes)

Image composition combines multiple visual elements into one image. Use Copy and Paste to bring elements from one image to another. The Scale tool resizes layers proportionally. The Crop tool trims the canvas to a specific area. The rule of thirds divides the image into a 3x3 grid—place important elements along the lines or intersections for a balanced, professional look.

Elaborate (8-10 minutes)

Think of arranging items on a shop display. You place the best products at eye level, group related items, and leave space for breathing. Image composition follows the same principles—balance, spacing, and focal points. When combining a portrait with a landscape background, where would you place the person using the rule of thirds?

Evaluate (5-8 minutes)

Exit ticket: Explain the rule of thirds in your own words. Then combine two images into one composition with at least three elements. If you finish early, add text labels to complete your collage.

Lesson 2.11 **Duration: 40 minutes**

Project - Photo Editing

Learning Objectives

- Apply all learned GIMP skills to edit a photo
- Practice selection, layers, filters, and text together
- Present a completed photo editing project

Engage (5-7 minutes)

Today is like your art exhibition day. You have learned to draw, paint, add effects, and compose images. Now you will use everything together to edit a real photograph—fixing colors, removing unwanted objects, adding text, and applying artistic effects. This is your chance to show all your skills.

Explore (10-12 minutes)

Choose one photograph to edit. Plan your workflow: what needs fixing? List at least four edits you will make (crop, color adjust, remove object, add text). Work through each step systematically, saving intermediate versions. Use layers to keep your edits organized.

Explain (10-12 minutes)

A complete photo editing workflow follows these steps: import the original, make basic adjustments (crop, brightness, contrast), apply creative edits (filters, effects), add text or graphics, and export the final image. Using layers throughout keeps your work non-destructive—you can always go back and change individual elements without starting over.

Elaborate (8-10 minutes)

Think of preparing for a school exhibition. You gather materials, make adjustments, add finishing touches, and display the final work. Photo editing follows the same process. Each skill you learned is a tool in your toolkit—today you use them all together. What story does your edited photo tell?

Evaluate (5-8 minutes)

Project submission: Present your edited photo with a brief description of the four or more edits you made. Explain which tools and techniques you used for each edit. If you finish early, create a second version with a different artistic style.

Lesson 2.12 **Duration: 40 minutes**

Chapter 2 Review

Learning Objectives

- Review all Chapter 2 GIMP concepts
- Connect graphics skills to real-world applications
- Identify areas needing further practice

Engage (5-7 minutes)

Think of a village craftsperson reviewing all the skills they learned—pottery, weaving, painting. Today we revisit all the GIMP skills we practiced: the interface, selection tools, layers, effects, drawing, color adjustments, filters, text, and composition. Let us see how much you have grown as a digital artist.

Explore (10-12 minutes)

Complete this GIMP matching quiz: match each tool or concept to its function. Work in pairs to finish as quickly as possible. Then trade papers with another pair and check their answers. Discuss any disagreements as a class.

Explain (10-12 minutes)

Chapter 2 covered image editing with GIMP: navigating the interface, selection tools for isolating areas, layers for organizing work, layer effects for depth, drawing tools for creation, color adjustments for correction, filters for artistic effects, text tools for typography, and composition for combining elements. Each skill builds on the previous ones to create complete digital artwork.

Elaborate (8-10 minutes)

Now that you know GIMP, think about your community. You could design posters for village events, edit photos for the school website, create greeting cards for festivals, or make educational materials. Name three real-world projects you could complete using your new GIMP skills.

Evaluate (5-8 minutes)

Chapter quiz: Answer ten questions covering all topics from Lesson 2.1 to 2.11. Include definitions, tool identification, and practical scenario questions. If you finish early, help a classmate with any difficult questions.

Lesson 3.1 **Duration: 40 minutes**

What is Multimedia?

Learning Objectives

- Define multimedia and its components
- Identify text, audio, image, video, and animation as media types
- Explain how multimedia is used in daily life

Engage (5-7 minutes)

Think of a village festival—there are decorations (images), drumbeats (audio), folk dances (video), and storytelling (text). When all these come together, it becomes a grand celebration. Multimedia works the same way on a computer, combining different media types.

Explore (10-12 minutes)

Work in groups of four. List examples of multimedia from your daily life—TV shows, phone videos, school presentations. Identify which media components (text, audio, image, video, animation) are present in each example. Share findings with the class.

Explain (10-12 minutes)

Multimedia is the combination of two or more media types to convey information. The five basic components are text, audio, images, video, and animation. Computers use multimedia in education, entertainment, and communication to make content engaging and interactive.

Elaborate (8-10 minutes)

A farmer in your village wants to create a mobile ad for his mangoes. Which media components would make his ad attractive? Write a plan listing the text, images, audio, and video he could include, and explain why each component helps sell his mangoes.

Evaluate (5-8 minutes)

Exit ticket: List the five components of multimedia. Give one real-life example where at least three components work together.

Lesson 3.2 **Duration: 40 minutes**

Audio Editing with Audacity

Learning Objectives

- Open Audacity and identify its interface components
- Record, play, and save a short audio clip
- Apply basic trimming and cutting operations on audio

Engage (5-7 minutes)

Think of a folk song singer in your village who recorded a song but sang extra lines by mistake. He needs to trim the unwanted parts. Audacity is like a digital scissors for audio—it lets you cut, trim, and clean up sound recordings.

Explore (10-12 minutes)

Open Audacity on the computer. Explore the toolbar buttons—Record, Play, Stop, Pause. Record a 10-second audio of yourself reading a sentence. Use the selection tool to highlight and delete an unwanted portion. Save your edited file.

Explain (10-12 minutes)

Audacity is a free audio editor. The record button captures sound through a microphone. The timeline shows your audio as a waveform. You can select parts of the waveform and cut, copy, or delete them. Always save your work in Audacity project format or export as MP3.

Elaborate (8-10 minutes)

Your school is preparing a bhajan recording for the annual day. Record yourself singing two lines of a bhajan. Trim any silence at the beginning and end. Practice using Undo (Ctrl+Z) when you make mistakes. Write down three keyboard shortcuts you discovered.

Evaluate (5-8 minutes)

Show your trimmed audio file to the teacher. Answer: What happens when you press Record while audio is already playing? Why is saving in project format important?

Lesson 3.3 **Duration: 40 minutes**

Audio Mixing

Learning Objectives

- Import multiple audio tracks into Audacity
- Adjust volume levels and pan left/right channels
- Mix multiple tracks and export the final audio

Engage (5-7 minutes)

Think of a temple festival where a tabla player, a flute player, and a singer perform together. The sound man adjusts each instrument's volume so they blend nicely. Audio mixing does the same thing digitally—combining different sounds into one balanced track.

Explore (10-12 minutes)

Open two audio files in Audacity as separate tracks. Use the gain slider on each track to make one louder and the other softer. Play both together and adjust until they sound balanced. Use the pan knob to move one track to the left speaker and the other to the right.

Explain (10-12 minutes)

Audio mixing combines multiple tracks into one. Each track has a gain slider for volume control and a pan knob for left-right balance. The mixer toolbar provides overall control. When satisfied, use File > Export > Export as MP3 to save the mixed audio.

Elaborate (8-10 minutes)

Create a short audio mix: record yourself speaking a poem on one track and import a saved instrumental clip on another track. Adjust volumes so the voice is clear over the music. Practice soloing and muting individual tracks. Write which combination sounded best.

Evaluate (5-8 minutes)

Demonstrate your mixed audio to the class. Answer: What is the difference between gain and pan? When would you use the mute button during mixing?

Lesson 3.4 **Duration: 40 minutes**

Video Concepts

Learning Objectives

- Define video and explain how frames create motion
- Identify video formats and their uses
- Describe the difference between analog and digital video

Engage (5-7 minutes)

Think of a flipbook you made as a child—flipping pages quickly made drawings appear to move. Video works the same way: hundreds of still images (frames) shown every second create the illusion of motion. Today we learn how video works on computers.

Explore (10-12 minutes)

Watch a short video on the computer. Right-click on the video file and check its properties to find the format (MP4, AVI, MKV). Compare file sizes of two videos with different resolutions. Discuss which format is commonly used on phones and why.

Explain (10-12 minutes)

Video is a sequence of images called frames displayed rapidly to create motion. Common formats include MP4 (most compatible), AVI (large files), and MKV (high quality). Frame rate (fps) determines smoothness—24 fps is cinema standard, 30 fps is common for web video.

Elaborate (8-10 minutes)

Your cousin records village sports events on a phone. Which video format should he use for sharing on WhatsApp versus keeping for a school project? Write your recommendations with reasons, considering file size and quality. Compare analog TV signals with digital video streaming.

Evaluate (5-8 minutes)

Answer: What is a frame? Why is MP4 the most commonly used video format? What happens to video quality when you decrease the frame rate?

Lesson 3.5 **Duration: 40 minutes**

Video Editing Basics

Learning Objectives

- Import video clips into a video editor
- Trim, split, and arrange clips on a timeline
- Add text overlays and export the edited video

Engage (5-7 minutes)

Think of a wedding videographer who films hours of ceremony but needs a 10-minute highlight video. He cuts unnecessary parts, arranges the best moments, and adds names as text. Video editing is this creative process done on a computer.

Explore (10-12 minutes)

Open a video editor (or use an online tool). Import two short video clips. Drag them to the timeline. Use the split tool to cut a clip in the middle. Remove unwanted sections. Arrange clips in order. Add a text title at the beginning saying "My Village."

Explain (10-12 minutes)

Video editing software provides a timeline where clips are arranged sequentially. The split tool cuts clips at the playhead position. Text overlays add titles and captions. After editing, export the video in a suitable format like MP4 with appropriate resolution for your needs.

Elaborate (8-10 minutes)

Record a 30-second video of your school compound. Import it into the editor. Trim the first and last 5 seconds. Add a title card with your name and class. Export the video and note the file size. Compare the edited file size with the original.

Evaluate (5-8 minutes)

Show your edited video to the class. Answer: What is the difference between trimming and splitting? Why should you save your project before exporting?

Lesson 3.6 **Duration: 40 minutes**

Presentation Software

Learning Objectives

- Identify features of presentation software like LibreOffice Impress
- Create a new presentation with multiple slides
- Add text, images, and shapes to slides

Engage (5-7 minutes)

Think of a teacher explaining a lesson using a chart on the blackboard. Presentation software is like a digital chart that can show text, pictures, and drawings on a screen—making lessons more colorful and interactive for students.

Explore (10-12 minutes)

Open LibreOffice Impress. Create a new presentation. Add three slides. On slide 1, add a title about your village. On slide 2, insert an image of a local landmark. On slide 3, draw a shape and add text inside it. Save the file.

Explain (10-12 minutes)

Presentation software like LibreOffice Impress creates slide-based presentations. Each slide can contain text boxes, images, shapes, charts, and multimedia. The slide panel on the left shows all slides. Use the toolbar to insert elements and format them with colors and fonts.

Elaborate (8-10 minutes)

Create a 5-slide presentation about "A Day in My Village." Include a title slide, slides about morning activities, school, evening sports, and a thank-you slide. Add at least one image and one shape. Practice using different slide layouts.

Evaluate (5-8 minutes)

Present your slides to the class. Answer: How many different slide layouts are available? What is the advantage of using a template over a blank presentation?

Lesson 3.7 **Duration: 40 minutes**

Slide Design

Learning Objectives

- Apply consistent themes and backgrounds to slides
- Use fonts, colors, and alignment for readability
- Create visually appealing slide layouts

Engage (5-7 minutes)

Think of how a rangoli artist uses colors, patterns, and symmetry to make a beautiful design at the entrance of a house. Slide design works similarly—using colors, fonts, and layout to make presentations attractive and easy to read.

Explore (10-12 minutes)

Open your village presentation. Apply a design theme from the Slide Properties panel. Change the background color. Adjust font sizes and colors for headings and body text. Ensure text is aligned properly. Check that all slides look consistent in style.

Explain (10-12 minutes)

Good slide design follows consistency—same fonts, colors, and layout across slides. Use high-contrast colors for readability. Headings should be larger than body text. Backgrounds should not distract from content. Limit text to key points and use the 6x6 rule (max 6 lines, 6 words each).

Elaborate (8-10 minutes)

Redesign your “Day in My Village” presentation. Apply a nature-themed background. Use one font family throughout. Ensure all headings are the same size and color. Add borders or frames to images. Write down three design rules you followed and why.

Evaluate (5-8 minutes)

Swap presentations with a partner and give design feedback. Answer: Why is consistency important in slide design? What makes text hard to read on a slide?

Lesson 3.8 **Duration: 40 minutes**

Animations

Learning Objectives

- Apply entrance, emphasis, and exit animations to objects
- Set animation order and timing
- Use animations effectively without overdoing them

Engage (5-7 minutes)

Think of a puppet show where characters enter the stage, perform actions, and exit. Animations in presentations work the same way—they control how objects appear, move, and disappear on a slide to draw attention to key points.

Explore (10-12 minutes)

Open your presentation. Select a text box and apply an entrance animation (like Fade In). Add an emphasis animation (like Spin) to an image. Apply an exit animation (like Fly Out) to a shape. Use the Animation panel to set the order and timing for each effect.

Explain (10-12 minutes)

Animations have three types: entrance (how objects appear), emphasis (how they draw attention while visible), and exit (how they disappear). The animation pane shows the sequence. You can set delay, duration, and trigger (on click or automatically). Less is more—use animations to enhance, not distract.

Elaborate (8-10 minutes)

Add animations to your village presentation. Make the title fade in first, bullet points appear one by one, and images zoom in. Set a 2-second delay between animations. Test the presentation in slideshow mode. Write which animation combinations looked professional.

Evaluate (5-8 minutes)

Present your animated slides. Answer: What is the difference between entrance and emphasis animations? Why should you avoid using too many different animations?

Lesson 3.9 **Duration: 40 minutes**

Slide Transitions

Learning Objectives

- Apply different transition effects between slides
- Set transition duration and advance timing
- Choose appropriate transitions for different contexts

Engage (5-7 minutes)

Think of turning pages in a storybook—sometimes you flip quickly, sometimes slowly, sometimes you fold the page. Slide transitions are like page turns in a digital book—they control how one slide moves to the next during a presentation.

Explore (10-12 minutes)

Open your presentation. Go to Slide Transition settings. Apply different transitions to each slide—Fade, Wipe, Push, Dissolve. Set the duration to 2 seconds for one and 0.5 seconds for another. Use "On Mouse Click" and "Automatically After" options. Preview in slideshow mode.

Explain (10-12 minutes)

Slide transitions control how the presentation moves from one slide to the next. Common transitions include Fade (smooth), Wipe (directional), and Push (slide in). Duration controls speed. You can set transitions to advance on mouse click or automatically after a set time for self-running presentations.

Elaborate (8-10 minutes)

Create a self-running presentation about "Types of Crops in Karnataka" that advances every 5 seconds. Use Fade for content slides and Dissolve for the title. Test the timing. Then create a 手动 version with Mouse Click transitions. Compare which version works better for different audiences.

Evaluate (5-8 minutes)

Demonstrate both versions to the class. Answer: When would you use automatic transitions? When is mouse-click better? What transition is most professional for a school project?

Lesson 3.10 **Duration: 40 minutes**

Multimedia Integration

Learning Objectives

- Embed audio and video clips into presentations
- Insert hyperlinks and interactive buttons
- Create a cohesive multimedia presentation

Engage (5-7 minutes)

Think of a science fair project where a student shows a chart, plays a recorded interview, and demonstrates a video—all from one computer screen. Multimedia integration combines all these elements into a single interactive presentation.

Explore (10-12 minutes)

Open your presentation. Insert an audio clip on one slide (Insert > Audio). Embed a video on another slide (Insert > Video). Add a hyperlink to a website on the title slide. Test all multimedia elements in slideshow mode. Ensure they play correctly.

Explain (10-12 minutes)

Multimedia integration means combining text, images, audio, video, and links in one presentation. Audio and video can be embedded or linked. Hyperlinks connect to websites or other slides. Interactive buttons create navigation menus. Test all elements before presenting to ensure everything works.

Elaborate (8-10 minutes)

Create a presentation about "Famous Temples of Karnataka." Include background music on the title slide, a video clip of a temple on another, hyperlinks to tourism websites, and navigation buttons. Ensure smooth playback. Write down any technical challenges you faced and how you solved them.

Evaluate (5-8 minutes)

Present your multimedia project. Answer: What is the difference between embedding and linking a video? Why is it important to test multimedia elements before presenting?

Lesson 3.11 **Duration: 40 minutes**

Project - Presentation

Learning Objectives

- Plan and design a complete multimedia presentation
- Apply all learned skills: design, animation, transitions, and multimedia
- Present and evaluate peer presentations effectively

Engage (5-7 minutes)

Think of a village fair where different stalls showcase their best products. Your presentation is your stall—it must attract visitors with good design, engaging content, and smooth delivery. Today you create your own showcase using everything you have learned.

Explore (10-12 minutes)

Choose a topic: “My Village Heritage,” “Festival Celebrations,” or “Local Heroes.” Plan 8-10 slides. Create a story outline. Gather images, record audio if needed, and write content. Build the presentation applying design principles, animations, transitions, and multimedia elements.

Explain (10-12 minutes)

A good project combines planning, design, and execution. Start with an outline, then create slides with consistent design. Add animations and transitions for flow. Include multimedia for engagement. Practice delivery timing. End with a conclusion slide. Peer feedback helps improve future projects.

Elaborate (8-10 minutes)

Complete your presentation project. Add a title slide, content slides with images and text, at least one audio or video element, smooth transitions, and a conclusion. Practice presenting in 5 minutes. Prepare to answer questions from classmates. Write a self-reflection on what you learned.

Evaluate (5-8 minutes)

Present your project to the class. Peer evaluation: Rate each presentation on design (1-5), content (1-5), multimedia use (1-5), and delivery (1-5). Provide one strength and one improvement suggestion for each presentation.

Lesson 3.12 **Duration: 40 minutes**

Chapter 3 Review

Learning Objectives

- Recall all multimedia concepts from Chapter 3
- Demonstrate audio and video editing skills
- Apply presentation design and multimedia integration

Engage (5-7 minutes)

Think of harvest season when farmers review their entire crop before selling. Today we review everything we learned about multimedia—from understanding components to creating complete presentations. This is our 知识收获 (knowledge harvest) time.

Explore (10-12 minutes)

Complete a hands-on challenge: Record and trim a 15-second audio clip, import it into a presentation, add 3 slides with images and text, apply animations and transitions, and present in 3 minutes. Time yourself and help classmates who need assistance.

Explain (10-12 minutes)

Chapter 3 covered multimedia components, audio editing with Audacity, audio mixing, video concepts, video editing basics, presentation software, slide design, animations, transitions, and multimedia integration. Each skill builds on the previous one to create complete multimedia projects.

Elaborate (8-10 minutes)

Create a mind map showing how all Chapter 3 topics connect. Start with “Multimedia” in the center and branch out to audio, video, and presentation topics. For each branch, write two key concepts and one real-life application. Compare your mind map with a partner’s.

Evaluate (5-8 minutes)

Quick quiz: Answer 10 questions covering all Chapter 3 topics. Practical test: Create a 3-slide presentation with image, text, animation, transition, and one multimedia element in 10 minutes. Score yourself honestly on areas that need more practice.

Lesson 4.1 **Duration: 40 minutes**

Introduction to Spreadsheets

Learning Objectives

- Define spreadsheet and its purpose in data management
- Identify rows, columns, cells, and cell references
- Open LibreOffice Calc and navigate the interface

Engage (5-7 minutes)

Think of a shopkeeper's ledger book where he writes items, quantities, and prices in neat rows and columns. A spreadsheet is a digital ledger—same rows and columns, but with the power to calculate, sort, and analyze data automatically.

Explore (10-12 minutes)

Open LibreOffice Calc. Explore the interface—identify the formula bar, cell reference box, column headers (A, B, C), and row numbers (1, 2, 3). Click on different cells and note their references. Type your name in cell A1 and a number in B1. Save the file.

Explain (10-12 minutes)

A spreadsheet is a grid of rows and columns used to organize and analyze data. Each intersection is a cell identified by its column letter and row number (like A1, B2). The formula bar shows cell contents. Spreadsheets are used for calculations, data analysis, charting, and record-keeping in schools, businesses, and farms.

Elaborate (8-10 minutes)

Create a simple inventory for your school library. In column A, list 5 book titles. In column B, write the number of copies. In column C, note the condition (Good/Fair). Use proper cell references for each entry. Count how many cells you used and write the total.

Evaluate (5-8 minutes)

Answer: What is a cell reference? Why are spreadsheets better than paper ledgers? Name three real-life uses of spreadsheets in your village.

Lesson 4.2 **Duration: 40 minutes**

Basic Formulas

Learning Objectives

- Write formulas using addition, subtraction, multiplication, and division
- Use the equals sign to start formulas
- Understand the difference between formula and value

Engage (5-7 minutes)

Think of a vegetable vendor who adds up daily sales manually-adding, subtracting, and multiplying on paper. Formulas in a spreadsheet do this instantly. Type the formula once, and the answer appears automatically. No more mental math errors!

Explore (10-12 minutes)

In a new spreadsheet, enter 100 in A1, 250 in B1, and 150 in C1. In D1, type =A1+B1 to add. In D2, type =B1-A1 to subtract. In D3, type =A1*2 to multiply. In D4, type =B1/5 to divide. Note how changing values in A1 automatically updates the formulas.

Explain (10-12 minutes)

Formulas always start with = sign. Basic operators are + (add), - (subtract), * (multiply), / (divide). Cell references (like A1) are used instead of numbers, so when data changes, results update automatically. This is the power of spreadsheets – dynamic calculations without rewriting formulas.

Elaborate (8-10 minutes)

Create a bill for a kirana store. List 5 items with prices in column A and quantities in column B. In column C, write formulas to calculate total for each item (=A1*B1). In cell C7, write a formula to find the grand total of all items. Change one price and observe the total update.

Evaluate (5-8 minutes)

Answer: Why do formulas start with =? What happens to C7 when you change a value in A1? Write the formula to calculate 15% tax on a total in cell D1.

Lesson 4.4 **Duration: 40 minutes**

Functions

Learning Objectives

- Use SUM, AVERAGE, COUNT, MAX, and MIN functions
- Enter functions using the function wizard
- Apply functions to real-world data sets

Engage (5-7 minutes)

Think of a teacher who needs to calculate the average marks of 40 students. Writing =A1+A2+…+A40 is tedious. Functions like SUM and AVERAGE do this in one step—they are shortcuts for common calculations that save hours of work.

Explore (10-12 minutes)

Enter marks for 5 subjects in A1:A5 (85, 90, 78, 92, 88). In B1, type =SUM(A1:A5) to find total. In B2, type =AVERAGE(A1:A5) for average. In B3, type =COUNT(A1:A5) to count entries. In B4, type =MAX(A1:A5) for highest. In B5, type =MIN(A1:A5) for lowest.

Explain (10-12 minutes)

Functions are built-in formulas for common calculations. SUM adds a range, AVERAGE finds the mean, COUNT counts numbers, MAX finds the largest, MIN finds the smallest. Use colon (:) to specify ranges like A1:A5. The function wizard (fx button) helps build complex functions step by step.

Elaborate (8-10 minutes)

Create a farmer's crop yield sheet. Enter yields for 6 months in B1:B6. Use SUM to find total annual yield, AVERAGE for monthly average, MAX for best month, MIN for worst month. Add a COUNT to verify all months are recorded. Format results with appropriate decimal places.

Evaluate (5-8 minutes)

Answer: What is the difference between SUM and AVERAGE? When would you use COUNT instead of SUM? Write a function to find the range (MAX minus MIN) of temperatures in C1:C7.

Lesson 4.5 **Duration: 40 minutes**

Charts Creation

Learning Objectives

- Select data ranges for chart creation
- Create bar, line, and pie charts
- Choose appropriate chart types for different data

Engage (5-7 minutes)

Think of a panchayat officer showing village development data—a bar chart of road construction, a pie chart of budget allocation, a line chart of population growth. Charts turn numbers into visual stories that anyone can understand at a glance.

Explore (10-12 minutes)

Enter monthly rainfall data: Months in A1:A6, Rainfall in mm in B1:B6. Select the data range. Go to Insert > Chart. Choose Bar Chart and click OK. Repeat to create a Pie Chart showing budget allocation and a Line Chart showing temperature changes over months.

Explain (10-12 minutes)

Charts visualize data for easier understanding. Bar charts compare quantities across categories. Line charts show trends over time. Pie charts show proportions of a whole. Select data first, then choose chart type. The chart wizard guides you through titles, labels, and legends.

Elaborate (8-10 minutes)

Create a chart for your class test scores: Names in A1:A10, Marks in B1:B10. Make a bar chart showing each student's marks. Then create a pie chart showing grade distribution (A, B, C grades). Add proper titles and labels. Which chart type better shows individual performance?

Evaluate (5-8 minutes)

Answer: When would you use a pie chart instead of a bar chart? What information does a legend provide? Create a line chart showing your study hours over a week and describe the trend.

Lesson 4.6	Duration: 40 minutes

Chart Formatting

Learning Objectives

- Modify chart titles, labels, and legends
- Change chart colors and styles
- Add trendlines and data labels to charts

Engage (5-7 minutes)

Think of a painter who first sketches a picture and then adds colors, shading, and details to make it beautiful. Chart formatting is like that—you create the basic chart first, then style it with colors, labels, and decorations to make it clear and attractive.

Explore (10-12 minutes)

Open your rainfall bar chart. Double-click to edit. Change the chart title to "Monthly Rainfall in Our Village." Add axis labels (Month, Rainfall in mm). Change bar colors to blue. Add data labels showing exact values. Try changing the chart style from the formatting options.

Explain (10-12 minutes)

Chart formatting improves readability and aesthetics. Titles explain what the chart shows. Axis labels define what each axis represents. Data labels show exact values on bars or points. Legends identify data series. Colors should be distinct and meaningful. Consistent formatting makes charts professional.

Elaborate (8-10 minutes)

Format your class marks bar chart. Add a meaningful title, axis labels, and data labels. Change colors to match school colors. Add a trendline to show the overall performance direction. Export the chart as an image. Compare your formatted chart with the original—list three improvements.

Evaluate (5-8 minutes)

Show your formatted chart to the class. Answer: Why are axis labels important? What makes a chart easy to read from a distance? How do data labels help when bars are similar in height?

Lesson 4.7 **Duration: 40 minutes**

Data Sorting

Learning Objectives

- Sort data in ascending and descending order
- Sort by single and multiple columns
- Use sort to organize records meaningfully

Engage (5-7 minutes)

Think of a library shelf where books are arranged by subject, then by author name within each subject. Sorting in spreadsheets works the same way—it organizes data so you can find information quickly, whether alphabetical, numerical, or by category.

Explore (10-12 minutes)

Enter student data: Names in A1:A8, Marks in B1:B8, Subjects in C1:C8. Sort by Marks (descending) to find toppers. Sort by Names (alphabetical) to organize the list. Then sort by Subjects first, then by Marks within each subject. Observe how the data rearranges.

Explain (10-12 minutes)

Sorting rearranges rows based on cell values. Ascending order is A-Z or 0-9; descending is Z-A or 9-0. Single-column sort orders by one column. Multi-column sort first orders by primary column, then by secondary column within groups. Always select all related columns before sorting to keep records intact.

Elaborate (8-10 minutes)

Create a village census spreadsheet: Names, Age, Occupation, Family Size. Sort by Occupation to group farmers together. Within farmers, sort by Age. Find the oldest and youngest farmer. Sort by Family Size to see largest families. Write which sorting helped answer which question.

Evaluate (5-8 minutes)

Answer: What happens if you sort only the Marks column without Names? When would you use multi-column sort? Sort your data by three different columns and describe what each sort reveals.

Lesson 4.8 **Duration: 40 minutes**

Data Filtering

Learning Objectives

- Apply auto-filter to display specific records
- Use filter criteria to show data meeting conditions
- Clear filters and understand filtered vs hidden data

Engage (5-7 minutes)

Think of a fisherman who uses a net with specific holes—small fish slip through, big fish stay. Data filtering works like a net—it shows only the records that match your criteria while hiding the rest. You choose what to catch and what to let go.

Explore (10-12 minutes)

Open your village census data. Apply Auto-Filter (Data > AutoFilter). Click the dropdown arrow on Occupation column. Select only "Farmer" to see only farmer records. Then filter by Age to show only people above 40. Clear filters one by one. Note how row numbers turn blue when filtered.

Explain (10-12 minutes)

Filtering temporarily hides data that doesn't match criteria—it doesn't delete anything. Auto-Filter adds dropdown arrows to column headers. You can filter by text (contains, equals), numbers (greater than, between), or dates. Multiple filters work together (AND logic). Clear filters to show all data again.

Elaborate (8-10 minutes)

Create a student database: Name, Class, Gender, Sports, Marks. Filter to show only girls who play cricket. Then filter to show Class 7 students with marks above 80. Apply three filters simultaneously. Clear filters and note how the total count changes. Write three questions that filtering can answer.

Evaluate (5-8 minutes)

Answer: What is the difference between filtering and deleting data? How do you know data is filtered? If you filter for "Class 7" AND "Marks > 80," which records appear?

Lesson 4.9 | Duration: 40 minutes

Data Validation

Learning Objectives

- Set validation rules for data entry in cells
- Create dropdown lists for restricted choices
- Display error messages for invalid entries

Engage (5-7 minutes)

Think of a temple bell that only rings when pulled correctly—if you push it, nothing happens. Data validation is like that—it prevents wrong data from being entered. You set rules, and the spreadsheet rejects entries that don't follow them.

Explore (10-12 minutes)

Select column B (Marks). Go to Data > Validity. Set criteria: Whole Number, between 0 and 100. Try entering 150—see the error message. Now create a dropdown list for Gender column with options: Male, Female, Other. Test the dropdown by selecting from the list.

Explain (10-12 minutes)

Data validation controls what users can enter in cells. You can restrict to numbers within a range, dates, text length, or create dropdown lists from predefined options. Custom error messages guide users when they enter wrong data. This prevents errors and maintains data quality in shared spreadsheets.

Elaborate (8-10 minutes)

Create a health record spreadsheet. Add validation: Age must be 5-100, Weight must be 20-100 kg, Blood Group must be from a dropdown (A+, B+, O+, AB+). Test each validation by trying invalid entries. Write custom error messages for each rule. Verify all validations work correctly.

Evaluate (5-8 minutes)

Answer: Why is data validation important in shared spreadsheets? Create a validation rule for phone numbers (10 digits). What happens when you paste data into validated cells?

Lesson 4.10 **Duration: 40 minutes**

Conditional Formatting

Learning Objectives

- Apply color scales to visualize data ranges
- Use data bars and icon sets for quick analysis
- Create custom rules for highlighting specific cells

Engage (5-7 minutes)

Think of a traffic light—green means go, yellow means slow, red means stop. Conditional formatting works like traffic lights for data—colors change automatically based on values, making it easy to spot patterns, problems, and highlights at a glance.

Explore (10-12 minutes)

Enter student marks in B1:B20. Select the range. Go to Format > Conditional Formatting. Apply a Color Scale: green for high marks, yellow for medium, red for low. Add Data Bars to see relative performance visually. Try Icon Sets (arrows or traffic lights). Observe how colors change with different values.

Explain (10-12 minutes)

Conditional formatting applies styles based on cell values. Color scales use gradients (green-yellow-red) to show value distribution. Data bars add colored bars proportional to values. Icon sets add symbols (arrows, flags) for categories. You can create custom rules like "highlight cells greater than 80" with specific colors.

Elaborate (8-10 minutes)

Apply conditional formatting to a temperature data set (30 values). Use color scale for hot-cold visualization. Create a rule to highlight temperatures above 40 in red. Add data bars to monthly rainfall data. Write which visualization method best shows extreme values and why.

Evaluate (5-8 minutes)

Answer: When is conditional formatting more useful than just reading numbers? Create a rule to highlight names starting with 'A' in blue. What is the difference between color scales and data bars?

Lesson 4.11 **Duration: 40 minutes**

Project - Grade Sheet

Learning Objectives

- Create a complete grade sheet using all spreadsheet skills
- Apply formulas, functions, charts, and formatting
- Analyze and present data findings effectively

Engage (5-7 minutes)

Think of a school headmaster preparing annual report cards—he needs marks, averages, ranks, and charts for each student. Today you become that headmaster, creating a professional grade sheet that combines everything you learned about spreadsheets.

Explore (10-12 minutes)

Create a grade sheet for 15 students. Columns: Roll No, Name, 5 subject marks, Total, Average, Grade, Rank. Use SUM for total, AVERAGE for average, nested IF for grades (A/B/C/D/F), and RANK for class position. Create charts for class average comparison and grade distribution.

Explain (10-12 minutes)

A grade sheet integrates all spreadsheet skills: data entry with validation, formulas for calculations, functions for analysis, conditional formatting for visual cues, and charts for presentation. The IF function creates grades based on conditions. RANK function assigns positions. Professional formatting makes the sheet ready for printing.

Elaborate (8-10 minutes)

Complete your grade sheet with all formulas working correctly. Add conditional formatting to highlight toppers (green) and failing students (red). Create a bar chart comparing student averages and a pie chart showing grade distribution. Format for printing with borders and headers. Write a summary of class performance.

Evaluate (5-8 minutes)

Present your grade sheet to the class. Answer: How did you handle students with equal marks in ranking? What grade boundaries did you use? How could this grade sheet save a teacher hours of work compared to manual calculation?

Lesson 4.12 **Duration: 40 minutes**

Chapter 4 Review

Learning Objectives

- Recall all data handling concepts from Chapter 4
- Demonstrate proficiency in formulas, functions, and charts
- Apply data analysis skills to solve real-world problems

Engage (5-7 minutes)

Think of a farmer at the end of harvest season who counts his produce, calculates profit, and plans for next season. Today we review everything about data handling—from entering data to analyzing it with charts. This is our data harvest review.

Explore (10-12 minutes)

Complete a data challenge: Enter raw data for 20 students (name, 4 subjects). Calculate totals and averages. Apply conditional formatting. Create a bar chart and pie chart. Sort by total. Filter to show only students scoring above 75. Use functions: SUM, AVERAGE, MAX, MIN, COUNT.

Explain (10-12 minutes)

Chapter 4 covered spreadsheet basics, formulas, cell references, functions, chart creation, chart formatting, data sorting, filtering, data validation, and conditional formatting. These skills work together to transform raw data into meaningful information for decision-making in schools, businesses, and daily life.

Elaborate (8-10 minutes)

Create a real-world project: Village health survey data for 20 families. Include family name, members, income, health issues, and distance to hospital. Analyze using all Chapter 4 skills. Write three insights your data reveals about the village. Create a presentation-ready report with charts.

Evaluate (5-8 minutes)

Quick quiz covering all Chapter 4 topics. Practical test: Given raw data, create a complete analysis with formulas, functions, sorting, filtering, and a chart in 15 minutes. Self-assess your strengths and areas needing more practice. Set goals for improving spreadsheet skills.

Lesson 5.1 **Duration: 40 minutes**

Scratch Review

Learning Objectives

- Recall Scratch interface components and their functions
- Identify sprites, costumes, and backdrops in projects
- Navigate the Scratch programming environment confidently

Engage (5-7 minutes)

Just as a farmer remembers the tools used in previous seasons, we revisit Scratch — the visual programming language that teaches coding through blocks. Like a familiar village square, Scratch provides a comfortable space where we've already learned the basics.

Explore (10-12 minutes)

Students open Scratch and explore the interface: stage, sprites, block palettes, and the scripting area. Like a farmer examining his plough before the monsoon, we check each component to ensure readiness for advanced programming ahead.

Explain (10-12 minutes)

Scratch is a block-based visual programming language developed by MIT. The interface includes the Stage (where programs run), Sprites (characters), Block Palette (commands organized by color), and Scripting Area (where blocks connect). Each block represents a command — like seeds that grow into actions.

Elaborate (8-10 minutes)

Students create a simple project by dragging blocks to make a sprite move, say "Namaskara," and change costumes. Like preparing soil before sowing, setting up our Scratch environment properly ensures our coding journey grows smoothly through this chapter.

Evaluate (5-8 minutes)

Can students identify and locate all Scratch interface components? Can they create a basic script using motion and looks blocks? Do they understand how sprites interact with the stage?

Lesson 5.2 | **Duration: 40 minutes**

Variables Advanced

Learning Objectives

- Create and use variables in Scratch projects
- Understand the difference between local and global variables
- Implement variables for score tracking and game mechanics

Engage (5-7 minutes)

In farming, a farmer tracks how many bags of rice were harvested or how much rain fell. Variables in Scratch work the same way — they store numbers and text that can change as the program runs, just like a farmer's harvest count increases through the season.

Explore (10-12 minutes)

Students experiment with creating variables: "score," "timer," and "lives." They see how variables appear as watchers on the stage, like a farmer's ledger showing current crop yield. Different blocks change variable values — set, change by, and show/hide.

Explain (10-12 minutes)

Variables are containers that store data. In Scratch, variables can hold numbers (like score) or text (like player name). Local variables exist only within a sprite, while global variables are accessible everywhere — like village-wide records versus individual farm records.

Elaborate (8-10 minutes)

Students build a simple counter: a sprite clicks to increase a score variable, then resets it. Like a shopkeeper counting daily sales, variables keep track of important numbers that change during program execution.

Evaluate (5-8 minutes)

Can students create variables and understand their purpose? Can they explain when to use local versus global variables? Do they successfully implement a score counter in their project?

Lesson 5.3	**Duration: 40 minutes**

Conditional Logic

Learning Objectives

- Understand if-then and if-then-else blocks in Scratch
- Create programs that make decisions based on conditions
- Apply conditional logic to real-world problem scenarios

Engage (5-7 minutes)

Every day, farmers make decisions: if it rains, irrigate the fields; otherwise, wait. If the crop price is high, sell; otherwise, store. Conditional logic in Scratch teaches computers to make similar if-then decisions, like a farmer's wisdom applied to programming.

Explore (10-12 minutes)

Students experiment with sensing blocks that return true or false. They connect conditions to action blocks, creating programs where sprites react differently based on touches, key presses, or variable values — like a farmer adjusting activities based on weather conditions.

Explain (10-12 minutes)

Conditional logic uses Boolean expressions (true/false) to determine actions. The "if-then" block executes code only when a condition is true. The "if-then-else" block offers two paths — one for true, one for false. Like a river splitting around a boulder, the program flows in different directions based on conditions.

Elaborate (8-10 minutes)

Students create a project where a sprite changes color when touching the mouse pointer. Like a chameleon changing colors based on surroundings, their sprite responds to environmental conditions using if-then blocks.

Evaluate (5-8 minutes)

Can students explain how if-then blocks work? Can they create programs with at least two different conditions? Do they understand Boolean logic concepts?

Lesson 5.4 **Duration: 40 minutes**

Loops Advanced

Learning Objectives

- Use repeat, forever, and repeat-until loops in Scratch
- Understand when to use different loop types
- Create animated sequences using loops effectively

Engage (5-7 minutes)

In a village, daily routines repeat: wake up, tend cattle, go to field, return home. Loops in Scratch automate repetition — like telling a computer to repeat an action many times without writing it again and again, just as a farmer's daily schedule repeats throughout the season.

Explore (10-12 minutes)

Students experiment with three loop types: "repeat" for a set number of times, "forever" for endless repetition, and "repeat-until" for condition-based repetition. Like a spinning potter's wheel, loops create continuous motion and action in their programs.

Explain (10-12 minutes)

The "repeat" block executes code a specified number of times. The "forever" block runs indefinitely until stopped. The "repeat-until" block continues until a condition becomes true. Like the cyclical nature of seasons — planting, growing, harvesting — loops create patterns that repeat with purpose.

Elaborate (8-10 minutes)

Students create a spinning animation using a repeat loop that changes sprite direction. Like a water wheel turning continuously, their sprite moves in repeated patterns, demonstrating how loops create efficient, elegant code.

Evaluate (5-8 minutes)

Can students differentiate between repeat, forever, and repeat-until loops? Can they create an animation that loops at least 10 times? Do they understand how loops reduce code repetition?

Lesson 5.5 **Duration: 40 minutes**

Sensing Blocks

Learning Objectives

- Explore Scratch sensing blocks for input detection
- Use distance, color, and key-press sensors in programs
- Create interactive projects that respond to user actions

Engage (5-7 minutes)

A farmer uses senses to understand the world: seeing if clouds gather, hearing if rain falls, feeling if the soil is moist. Scratch sensing blocks give sprites similar abilities — to detect touches, colors, distances, and key presses, like digital senses for virtual characters.

Explore (10-12 minutes)

Students test sensing blocks: touching mouse-pointer, touching color, distance to sprite, and key pressed. Each block returns true or false, like a farmer's yes/no decisions based on observations. They see how sensing blocks connect to conditional logic.

Explain (10-12 minutes)

Sensing blocks detect the program's environment. "Touching?" checks if a sprite contacts something. "Distance to" measures how far apart objects are. "Key pressed?" detects keyboard input. "Color touching?" checks color contact. These blocks are the computer's sensory organs, providing information about the world around them.

Elaborate (8-10 minutes)

Students create a game where a sprite follows the mouse pointer, changing color when it touches the edge. Like a goat following its shepherd, the sprite uses sensing to stay connected to user input, creating responsive interaction.

Evaluate (5-8 minutes)

Can students use at least three different sensing blocks? Can they create a project where sprites respond to keyboard input? Do they understand how sensing blocks provide program input?

Lesson 5.6 **Duration: 40 minutes**

Sprite Interactions

Learning Objectives

- Create multiple sprites that interact with each other
- Use broadcasts to coordinate sprite communication
- Design scenarios where sprites work together to complete tasks

Engage (5-7 minutes)

In a village, people work together: the farmer ploughs, the wife sows, children chase birds. Each has a role, and they coordinate through words and signals. Sprite interactions in Scratch teach how multiple characters communicate and work together, like villagers coordinating during harvest season.

Explore (10-12 minutes)

Students create two sprites and explore how they can interact: following each other, changing when they touch, or responding to the same events. Like farmers in adjacent fields communicating across boundaries, sprites share information through broadcasts and direct sensing.

Explain (10-12 minutes)

Sprites interact through several methods: direct sensing (touching another sprite), broadcasts (messages sent between sprites), and shared variables (common data). Broadcasts are like village drums — one sprite sends a message, and all sprites listening respond accordingly, coordinating their actions.

Elaborate (8-10 minutes)

Students create a simple scene where a cat sprite chases a mouse sprite. When they touch, both sprites broadcast messages to change their costumes. Like a chase game in the village square, sprites interact through movement, sensing, and communication.

Evaluate (5-8 minutes)

Can students create at least two sprites that interact? Can they implement broadcast communication between sprites? Do they understand how sprites can coordinate their actions?

Lesson 5.7 **Duration: 40 minutes**

Clones

Learning Objectives

- Understand what clones are and how they work in Scratch
- Create and delete clones dynamically during program execution
- Use clones to build multi-character game scenarios

Engage (5-7 minutes)

In farming, a farmer plants many identical seeds — each grows into the same type of plant. Clones in Scratch work similarly: one original sprite creates copies of itself, like a farmer sowing multiple seeds that all grow into the same crop, each clone identical to the parent.

Explore (10-12 minutes)

Students create a sprite and use "create clone" to make copies. They see each clone acts independently but shares the same costumes and scripts. Like identical twins in the village, clones look the same but can move and behave differently based on their own variables.

Explain (10-12 minutes)

Clones are temporary copies of a sprite that appear during program execution. When a clone is created, it receives the parent's scripts, costumes, and sounds. Clones can be deleted when no longer needed. Like harvesting individual plants from a field, clones appear, perform their purpose, and can be removed.

Elaborate (8-10 minutes)

Students create a project where clicking creates clones that fall like raindrops. Each clone falls independently, creating a rain effect. Like monsoon droplets from clouds, clones multiply from one source and perform identical actions with slight variations.

Evaluate (5-8 minutes)

Can students create clones from an original sprite? Can they make clones behave independently? Do they understand when and why to use clones in a project?

Lesson 5.8 | **Duration: 40 minutes**

Game Design

Learning Objectives

- Understand core game mechanics: objectives, rules, and feedback
- Apply game design principles to create engaging gameplay
- Balance difficulty and reward in game elements

Engage (5-7 minutes)

Children in villages create games from available materials: marbles, hide-and-seek, kite flying. Each game has rules, objectives, and fun. Game design in Scratch teaches how to create digital games with the same principles — clear goals, fair rules, and satisfying rewards, like village games translated to computer screens.

Explore (10-12 minutes)

Students analyze popular games: what makes them fun? They identify core elements: objectives (what to achieve), rules (how to play), challenges (obstacles), and feedback (score, sounds). Like a village 游戏 master explaining rules, they learn to structure engaging experiences.

Explain (10-12 minutes)

Good game design includes: clear objectives (collect 10 coins), fair rules (don't touch enemies), progressive challenges (increasing difficulty), and immediate feedback (score increases, sounds play). Like a well-organized village sports day, games need structure to be enjoyable and challenging.

Elaborate (8-10 minutes)

Students design a simple game concept on paper: objective, rules, characters, and challenges. Like planning a village festival, they think about what will attract participants, keep them engaged, and make them want to play again.

Evaluate (5-8 minutes)

Can students identify the four core elements of game design? Can they create a basic game concept document? Do they understand how game mechanics create engaging experiences?

Lesson 5.9 **Duration: 40 minutes**

Keyboard Controls

Learning Objectives

- Implement keyboard input controls for sprite movement
- Create responsive controls using arrow keys and other inputs
- Design smooth movement systems for game characters

Engage (5-7 minutes)

When driving a bullock cart, the farmer controls direction with the reins — left, right, forward. Keyboard controls in Scratch work the same way: arrow keys steer sprites like reins steer cattle, giving players control over character movement in games and animations.

Explore (10-12 minutes)

Students experiment with "when key pressed" blocks for all four arrow keys. They adjust movement amounts and directions, seeing how responsive controls feel. Like a farmer learning to guide oxen smoothly, they fine-tune controls for natural, responsive movement.

Explain (10-12 minutes)

Keyboard controls use event blocks ("when key pressed") to trigger movement. Each arrow key maps to a direction: up (y+), down (y-), left (x-), right (x+). The change amount determines speed — small values for precise control, large values for quick movement. Like adjusting reins for different terrains, control sensitivity affects gameplay feel.

Elaborate (8-10 minutes)

Students create a character that moves smoothly with arrow keys across a farm-themed backdrop. They add boundaries to prevent leaving the stage, like fences keeping cattle in the field, ensuring controlled, purposeful movement.

Evaluate (5-8 minutes)

Can students implement all four directional controls? Do their controls feel responsive and smooth? Can they add boundaries to constrain movement?

Lesson 5.10 **Duration: 40 minutes**

Random Events

Learning Objectives

- Use random number generation in Scratch programs
- Create unpredictable game elements using randomness
- Understand how randomness adds variety and replayability

Engage (5-7 minutes)

Nature is unpredictable — rain may fall unexpectedly, a cow may wander to a different field, a bird may land anywhere. Random events in Scratch bring this unpredictability to programs, making games more interesting like the natural variety farmers experience daily.

Explore (10-12 minutes)

Students experiment with “pick random” blocks for positions, directions, and values. They see how random numbers create varied outcomes each time the program runs. Like dice rolling during village gatherings, randomness brings chance and surprise to their programs.

Explain (10-12 minutes)

The “pick random” block generates numbers within a specified range. Using random positions makes sprites appear unpredictably. Random directions create varied movement patterns. Random timing adds suspense. Like the unpredictable monsoon rains that bring both joy and challenge, randomness makes programs dynamic and engaging.

Elaborate (8-10 minutes)

Students create a game where coins appear at random positions for the player to collect. Each playthrough is different because positions are randomized, like harvesting different amounts of produce each season — never exactly the same twice.

Evaluate (5-8 minutes)

Can students implement random number generation in their projects? Can they explain why randomness improves games? Do they use random values in at least three different ways?

Lesson 5.11 **Duration: 40 minutes**

Project - Game

Learning Objectives

- Apply all learned Scratch concepts to create a complete game
- Integrate variables, loops, conditionals, and sprite interactions
- Document and present a finished game project to peers

Engage (5-7 minutes)

Like farmers bringing their best harvest to the village market, students now showcase everything they've learned by creating a complete game. This project combines all the tools — variables, loops, conditionals, sprites — like combining seeds, water, and sunlight to grow a bountiful crop.

Explore (10-12 minutes)

Students plan their game: choose a theme (farming, village life, or adventure), design objectives, create sprites, and plan interactions. Like farmers planning their fields before monsoon, careful planning ensures successful game creation.

Explain (10-12 minutes)

A complete game requires: planning (design document), implementation (coding sprites and scripts), testing (finding and fixing bugs), and presentation (sharing with others). Like a farmer's complete journey from ploughing to harvesting, game development follows a structured process from concept to completion.

Elaborate (8-10 minutes)

Students work in pairs to create their games, dividing tasks: one designs sprites, the other codes scripts. Like farming partnerships where labor is shared, collaboration produces better results than working alone.

Evaluate (5-8 minutes)

Does the game include at least three different Scratch concepts learned? Is the game playable and enjoyable? Can students explain their coding choices and demonstrate their game to classmates?

Lesson 5.12 **Duration: 40 minutes**

Chapter 5 Review

Learning Objectives

- Review and consolidate all Scratch concepts from Chapter 5
- Demonstrate mastery through a comprehensive review activity
- Identify areas for further practice and improvement

Engage (5-7 minutes)

Like a farmer reviewing the season's work before the next planting, we look back at everything learned in Chapter 5. From basic Scratch review to complete game creation, we've built skills step by step — like terrace farming where each level supports the next.

Explore (10-12 minutes)

Students complete a review challenge: create a mini-project that uses variables, loops, conditionals, sensing, clones, and keyboard controls. Like a farmer demonstrating all farming skills during harvest, this activity shows mastery of all concepts.

Explain (10-12 minutes)

Chapter 5 covered Scratch fundamentals: Review (5.1), Variables (5.2), Conditional Logic (5.3), Loops (5.4), Sensing (5.5), Sprite Interactions (5.6), Clones (5.7), Game Design (5.8), Keyboard Controls (5.9), Random Events (5.10), Project (5.11), and Review (5.12). Like farming techniques from ploughing to harvesting, each concept builds upon previous knowledge.

Elaborate (8-10 minutes)

Students reflect on their learning journey: what was easiest, what was hardest, and what they want to learn more about. Like farmers planning improvements for next season, self-reflection guides future learning goals.

Evaluate (5-8 minutes)

Can students demonstrate all major Scratch concepts in their review project? Can they explain each concept's purpose and usage? Have they identified areas for continued practice?

Lesson 6.1 **Duration: 40 minutes**

What is Digital Citizenship?

Learning Objectives

- Define digital citizenship and its importance in modern life
- Identify the core principles of being a responsible digital citizen
- Connect digital citizenship concepts to real-life village community values

Engage (5-7 minutes)

In our village, we follow unwritten rules: respect elders, help neighbors, keep promises. Digital citizenship applies these same values to the online world — how we behave, treat others, and use technology responsibly, just like being a good member of our village community.

Explore (10-12 minutes)

Students discuss what makes someone a good citizen in their village: honesty, respect, helping others. They compare these to online behavior — being honest on social media, respecting others' posts, and reporting harmful content, like a digital version of village community values.

Explain (10-12 minutes)

Digital citizenship means using technology responsibly, ethically, and safely. It includes five core principles: respect (treating others well online), educate (learning about technology), protect (keeping yourself and others safe), responsibility (accountability for actions), and etiquette (polite online behavior). Like village values that guide community life, these principles guide our digital lives.

Elaborate (8-10 minutes)

Students create a "Digital Village Pledge" — rules they'll follow online, based on village values. Like a village promise that everyone agrees to keep, their pledge commits them to responsible digital behavior.

Evaluate (5-8 minutes)

Can students define digital citizenship in their own words? Can they identify at least three core principles? Do they see the connection between village values and digital behavior?

Lesson 6.2 **Duration: 40 minutes**

Online Safety Rules

Learning Objectives

- Learn essential online safety rules for internet use
- Identify common online dangers and how to avoid them
- Create a personal online safety checklist

Engage (5-7 minutes)

Before a farmer goes to the market, he takes precautions: tells family where he's going, carries money safely, watches for strangers. Online safety rules are similar — before going on the internet, we need precautions to protect ourselves from digital dangers, just like farmers protecting themselves in unfamiliar places.

Explore (10-12 minutes)

Students brainstorm online dangers: strangers asking personal information, fake websites, harmful downloads, and cyberbullying. Like recognizing thorny plants in the forest, they learn to identify digital threats that could harm them.

Explain (10-12 minutes)

Key online safety rules: never share personal information (name, address, school, phone), use strong passwords, only visit trusted websites, tell a trusted adult if something feels wrong, and never meet online strangers in person. Like safety rules for working with farm equipment, these rules prevent injury in the digital world.

Elaborate (8-10 minutes)

Students create a personal safety checklist with at least five rules, illustrated with drawings. Like a farmer's checklist before monsoon preparation, their digital safety checklist becomes a reference for safe internet use.

Evaluate (5-8 minutes)

Can students list at least five online safety rules? Can they explain why each rule is important? Do they have a written safety checklist they can follow?

Lesson 6.3 **Duration: 40 minutes**

Password Security

Learning Objectives

- Understand why passwords are important for online security
- Learn to create strong, memorable passwords
- Practice good password management habits

Engage (5-7 minutes)

A farmer locks his granary to protect the harvest - the lock keeps strangers out. A password is like a digital lock that protects your online accounts. Without a strong password, anyone can enter your digital granary and steal your information, like thieves entering an unlocked store.

Explore (10-12 minutes)

Students evaluate password strength: "password" (weak), "rajan123" (weak), "BlueMango42!" (strong). Like comparing different locks - a paper lock versus a steel lock - they see how password complexity affects security.

Explain (10-12 minutes)

Strong passwords include: at least 12 characters, mix of uppercase and lowercase letters, numbers, and special characters (`!@#$%`). Avoid personal information (birthdays, names). Use different passwords for different accounts. Like using different locks for different doors, unique passwords protect each account separately.

Elaborate (8-10 minutes)

Students create strong passwords for imaginary accounts, following the rules. They also practice the "password manager" concept - writing passwords in a secure place (not on paper under the keyboard). Like a farmer's secret storage location, passwords need safe keeping.

Evaluate (5-8 minutes)

Can students create passwords meeting all strength criteria? Can they explain why personal information shouldn't be in passwords? Do they understand the importance of unique passwords for different accounts?

Lesson 6.4 **Duration: 40 minutes**

Cyberbullying

Learning Objectives

- Define cyberbullying and recognize its various forms
- Understand the emotional and social impact of cyberbullying
- Learn strategies to prevent and respond to cyberbullying

Engage (5-7 minutes)

In villages, we sometimes hear unkind words about others — but online, hurtful messages can spread to hundreds instantly. Cyberbullying is using technology to hurt, embarrass, or threaten others, and it's as serious as bullying in the village square, just harder to escape.

Explore (10-12 minutes)

Students identify different forms of cyberbullying: mean messages, spreading rumors online, sharing embarrassing photos, excluding someone from online groups, and impersonation. Like recognizing different types of harassment in village life, they learn to identify digital bullying patterns.

Explain (10-12 minutes)

Cyberbullying includes: sending hurtful messages, spreading rumors, sharing private information, creating fake profiles, and excluding others. Unlike village bullying that ends when you go home, cyberbullying follows you everywhere through phones and computers. Like a storm that follows you across fields, it's harder to escape.

Elaborate (8-10 minutes)

Students role-play scenarios: what to do if you're cyberbullied (tell an adult, don't respond, save evidence), if you see it happening (report, support the victim), and how to prevent it (think before posting, treat others as you want to be treated). Like village elders mediating conflicts, they learn peaceful resolution strategies.

Evaluate (5-8 minutes)

Can students define cyberbullying and give examples? Can they identify at least three response strategies? Do they understand the importance of reporting cyberbullying to trusted adults?

Lesson 6.5 **Duration: 40 minutes**

Digital Footprint

Learning Objectives

- Understand what a digital footprint is and how it's created
- Recognize the permanence of online actions and their consequences
- Learn to manage and protect their digital footprint

Engage (5-7 minutes)

When walking on muddy ground, we leave footprints that others can follow. Online, every action leaves a digital footprint — posts, photos, comments, searches — that can be found and followed by others, like footprints in wet earth that tell where we've been.

Explore (10-12 minutes)

Students list actions that create digital footprints: social media posts, online searches, app downloads, comments on videos, and email messages. Like a farmer's daily path through fields, each online step leaves a trace that can be followed.

Explain (10-12 minutes)

A digital footprint is the trail of data left by online activities. It includes active footprints (things you intentionally share) and passive footprints (data collected without your knowledge). Like footprints in sand that remain until washed away by waves, digital footprints can last indefinitely on the internet.

Elaborate (8-10 minutes)

Students audit their own digital footprint: what would someone find if they searched their name? They create a plan to manage their footprint positively — sharing good things, avoiding harmful content, and cleaning up old posts. Like a farmer maintaining clean paths for others to follow.

Evaluate (5-8 minutes)

Can students explain what a digital footprint is? Can they identify at least three types of online actions that create footprints? Do they have a plan to manage their digital presence positively?

Lesson 6.6 **Duration: 40 minutes**

Online Predators

Learning Objectives

- Understand who online predators are and how they target children
- Recognize warning signs of predatory behavior online
- Learn protective strategies to stay safe from online predators

Engage (5-7 minutes)

In the village, we teach children not to take sweets from strangers. Online predators are like digital strangers who try to gain children's trust for harmful purposes. They pretend to be friendly while hiding their true intentions, like a wolf in sheep's clothing.

Explore (10-12 minutes)

Students identify warning signs: adults asking personal questions, requesting photos, wanting to meet in person, offering gifts, or asking to keep secrets. Like recognizing suspicious behavior in unfamiliar village visitors, these signs indicate potential danger.

Explain (10-12 minutes)

Online predators use tactics: building trust through compliments, offering gifts or game currency, asking to move to private messaging, requesting photos, and isolating children from friends and family. Like farmers protecting livestock from predators, children must recognize and avoid these dangerous tactics.

Elaborate (8-10 minutes)

Students practice safe responses: never share personal information, don't respond to suspicious messages, tell a trusted adult immediately, and never agree to meet online contacts. Like a village safety drill, practicing these responses builds protective habits.

Evaluate (5-8 minutes)

Can students identify at least three warning signs of online predators? Can they explain why adults shouldn't ask children for personal information? Do they know who to tell if they encounter suspicious online behavior?

Lesson 6.7 **Duration: 40 minutes**

Copyright Basics

Learning Objectives

- Understand what copyright is and why it exists
- Recognize copyright symbols and their meaning
- Learn how to use copyrighted material legally

Engage (5-7 minutes)

A farmer's seeds are his own — others can't take them without permission. Copyright works the same way for creative work: books, music, art, and code belong to their creators, and others must ask permission to use them, like respecting a farmer's ownership of his harvest.

Explore (10-12 minutes)

Students identify copyrighted materials around them: textbooks, songs, movies, and software. They see the © symbol and learn it means "this work belongs to someone." Like recognizing property boundaries in the village, copyright marks creative ownership.

Explain (10-12 minutes)

Copyright gives creators exclusive rights to their work: to copy, distribute, perform, and modify it. The © symbol indicates copyright protection. Copyright exists automatically when work is created — no registration needed. Like a farmer's claim to his field, copyright protects creators' rights to their creative produce.

Elaborate (8-10 minutes)

Students learn about fair use: using small portions for education, criticism, or parody. Like borrowing a neighbor's tool for a day (acceptable) versus taking it permanently (not acceptable), fair use allows limited borrowing while respecting ownership.

Evaluate (5-8 minutes)

Can students explain what copyright protects? Can they identify the © symbol and its meaning? Do they understand the difference between using copyrighted material legally versus illegally?

Lesson 6.8 **Duration: 40 minutes**

Plagiarism

Learning Objectives

- Define plagiarism and understand why it's wrong
- Learn proper ways to credit original sources
- Practice paraphrasing and citing sources correctly

Engage (5-7 minutes)

If a student copies another's homework and claims it as their own, it's cheating — plagiarism is the same for any work. Using someone else's words, ideas, or work without credit is like taking credit for a neighbor's harvest — dishonest and disrespectful.

Explore (10-12 minutes)

Students identify plagiarism examples: copying homework, using internet images without credit, presenting someone else's project as their own, and paraphrasing without citation. Like recognizing different forms of theft in village life, they learn to identify academic dishonesty.

Explain (10-12 minutes)

Plagiarism is using others' work without attribution. It includes: copying text word-for-word, using ideas without credit, submitting purchased work, and self-plagiarism (reusing your own work). Like a farmer taking credit for another's harvest, plagiarism steals others' creative labor.

Elaborate (8-10 minutes)

Students practice proper citation: copying a sentence and adding quotation marks, paraphrasing information and crediting the source, and creating a simple reference list. Like recording which field seeds came from, proper citation respects source origins.

Evaluate (5-8 minutes)

Can students define plagiarism in their own words? Can they demonstrate at least two proper citation methods? Do they understand why plagiarism is harmful to both original creators and themselves?

Lesson 6.9 | Duration: 40 minutes

Phishing Scams

Learning Objectives

- Understand what phishing is and how it works
- Identify common phishing tactics and warning signs
- Learn how to protect themselves from phishing attempts

Engage (5-7 minutes)

A fisherman uses bait to catch fish — phishing works the same way online. Scammers send fake emails, messages, or websites as bait to steal personal information like passwords and bank details. Like a trap disguised as food, phishing lures victims with false promises.

Explore (10-12 minutes)

Students examine example phishing messages: fake prize notifications, urgent bank alerts, and suspicious login requests. They identify warning signs: misspellings, urgent language, suspicious links, and requests for personal information. Like recognizing fake currency in the village market, they learn to spot deceptive messages.

Explain (10-12 minutes)

Phishing uses deception to steal information. Common tactics: fake emails claiming account problems, messages about prizes or gifts, urgent requests for password verification, and links to fake websites. Like a stranger offering "free" gifts that require personal details, phishing hides theft behind generosity.

Elaborate (8-10 minutes)

Students practice phishing detection: checking sender addresses, hovering over links before clicking, verifying requests through official channels, and reporting suspicious messages. Like a farmer testing seeds before planting, verification prevents costly mistakes.

Evaluate (5-8 minutes)

Can students identify at least three warning signs of phishing? Can they explain why clicking suspicious links is dangerous? Do they know the proper steps to take when receiving a suspicious message?

Lesson 6.10 **Duration: 40 minutes**

Screen Time Management

Learning Objectives

- Understand the effects of excessive screen time on health
- Learn strategies for balanced technology use
- Create a personal screen time management plan

Engage (5-7 minutes)

A farmer knows when to work in the field and when to rest — too much work without rest harms health. Similarly, too much screen time without breaks harms our eyes, posture, and mental health. Balance is key, like a farmer balancing work and rest through the seasons.

Explore (10-12 minutes)

Students track their typical screen time: how many hours daily for schoolwork, entertainment, and socializing. They identify activities that could replace some screen time: outdoor games, reading, helping at home, and family time. Like a farmer rotating crops to keep soil healthy, they learn to rotate activities.

Explain (10-12 minutes)

Excessive screen time can cause: eye strain, poor posture, sleep problems, reduced physical activity, and social isolation. Recommended limits: no more than 2 hours of recreational screen time daily, with breaks every 20 minutes. Like irrigation schedules that prevent waterlogging, screen time limits prevent health problems.

Elaborate (8-10 minutes)

Students create a weekly schedule that balances screen time with physical activity, study, family time, and sleep. Like a farmer's balanced daily routine, their schedule ensures healthy technology use alongside other important activities.

Evaluate (5-8 minutes)

Can students identify at least three negative effects of excessive screen time? Can they create a realistic weekly schedule balancing screen time with other activities? Do they understand the importance of regular breaks during screen use?

Lesson 6.11 **Duration: 40 minutes**

Project - Safety Poster

Learning Objectives

- Apply digital citizenship knowledge to create an educational poster
- Design visual content that communicates safety messages effectively
- Present and explain their poster to classmates

Engage (5-7 minutes)

Like village awareness campaigns about health or farming practices, students now create posters teaching others about digital safety. A well-designed poster can spread important messages across the community, like a village bulletin board that everyone reads.

Explore (10-12 minutes)

Students choose a digital citizenship topic from Chapter 6 to illustrate. They research statistics, collect examples, and plan visual layouts. Like village health workers planning awareness campaigns, they consider their audience and the most effective messages.

Explain (10-12 minutes)

Effective posters include: clear title, eye-catching visuals, key statistics or facts, simple language, and call-to-action. Like village announcements that must grab attention quickly, digital safety posters need to communicate messages at a glance while providing enough detail to inform.

Elaborate (8-10 minutes)

Students create their posters using paper, markers, and printed images. They practice presenting: explaining their topic, why it's important, and what viewers should do. Like farmers sharing knowledge at village meetings, they become digital safety educators.

Evaluate (5-8 minutes)

Does the poster clearly communicate one digital citizenship message? Is the poster visually appealing and easy to understand? Can students present their poster and answer questions about the topic?

Lesson 6.12 **Duration: 40 minutes**

Chapter 6 Review

Learning Objectives

- Review and consolidate all digital citizenship concepts from Chapter 6
- Demonstrate understanding through a comprehensive review activity
- Create a personal action plan for responsible digital citizenship

Engage (5-7 minutes)

Like a farmer reviewing the season's harvest before planning next year, we reflect on everything learned about digital citizenship in Chapter 6. From basic online safety to creating educational posters, we've built knowledge step by step — like terrace farming where each concept supports the next.

Explore (10-12 minutes)

Students complete a digital citizenship quiz covering all chapter topics. They identify areas where they feel confident and areas needing more practice. Like farmers assessing which crops thrived and which need improvement, they evaluate their learning progress.

Explain (10-12 minutes)

Chapter 6 covered: Digital Citizenship (6.1), Online Safety (6.2), Passwords (6.3), Cyberbullying (6.4), Digital Footprint (6.5), Online Predators (6.6), Copyright (6.7), Plagiarism (6.8), Phishing (6.9), Screen Time (6.10), Project (6.11), and Review (6.12). Like a farmer's complete knowledge cycle from planning to harvest, these concepts prepare students for safe, responsible technology use.

Elaborate (8-10 minutes)

Students create a personal "Digital Citizenship Action Plan" — specific commitments for online behavior. Like a farmer's resolution for the next season, their plan includes concrete actions for being responsible digital citizens.

Evaluate (5-8 minutes)

Can students demonstrate knowledge of all major digital citizenship concepts? Can they apply these concepts to real-life scenarios? Do they have a personal action plan for responsible technology use?